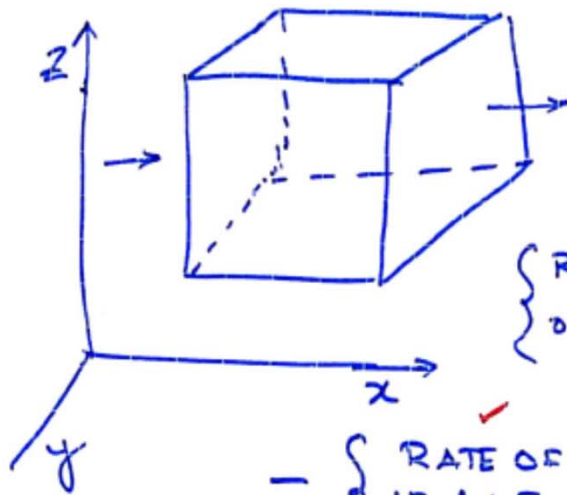


# **Heat Transfer Fundamentals**

## Energy Equation

EQUATION OF CHANGE FOR A NON ISOTHERMAL SYSTEM

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FLUID IN <sup>THRU</sup>  $x, y, z$  FACES  
 $\Delta x \Delta y, \Delta y \Delta z, \Delta z \Delta x$   
 FLUID OUT "  $x+\Delta x, y+\Delta y, z+\Delta z$

CONSERVATION OF ~~ENG~~ ENERGY.

$$\begin{aligned}
 & \left\{ \begin{array}{l} \text{RATE OF ACCUMULATION} \\ \text{OF } \underline{\text{INTERNAL \& KINETIC}} \\ \text{ENERGY} \end{array} \right\} = \left\{ \begin{array}{l} \text{RATE OF} \\ \text{IE \& KE} \\ \text{IN} \\ \text{BY } \underline{\text{CONV.}} \end{array} \right\} \\
 & - \left\{ \begin{array}{l} \text{RATE OF} \\ \text{IE \& KE} \\ \text{OUT} \\ \text{BY } \underline{\text{CONV.}} \end{array} \right\} + \left\{ \begin{array}{l} \underline{\text{NET}} \text{ RATE} \\ \text{OF HEAT ADDITION} \\ \text{BY } \underline{\text{COND.}} \end{array} \right\} - \left\{ \begin{array}{l} \text{NET RATE OF} \\ \text{WORK DONE} \\ \text{BY THE SYSTEM} \\ \text{ON THE} \\ \text{SURROUNDINGS} \end{array} \right\} \\
 & - \text{1st LAW OF THERMODYNAMICS FOR} \\
 & \quad \text{AN } \underline{\text{OPEN}} \text{ SYSTEM}
 \end{aligned}$$

$$\checkmark \text{ RATE OF ACCUM OF IE + KE} = \Delta x \Delta y \Delta z \frac{\partial}{\partial t} \left( \rho U + \frac{1}{2} \rho v^2 \right)$$

$U =$  IE PER UNIT MASS OF THE FLUID

$v =$  MAG. OF LOCAL FLUID VEL.

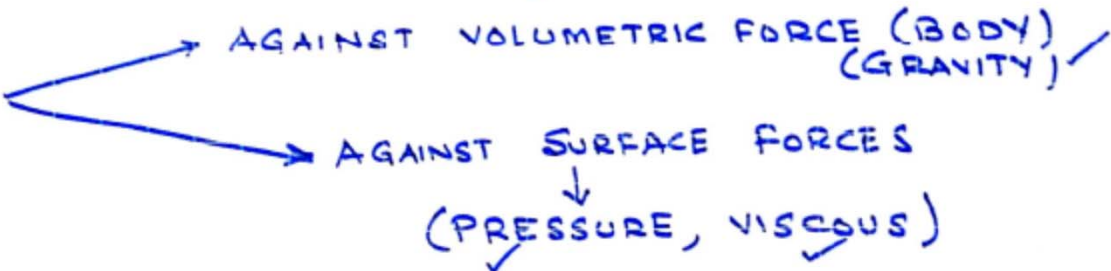
RATE OF CONVECTION OF IE & KE INTO THE ELEMENT

$$\begin{aligned} & \overset{(x \text{ FACE})}{\Delta y \Delta z} \left\{ \overset{\text{IN}}{v_x} \left( \rho U + \frac{1}{2} \rho v^2 \right) \Big|_x - \overset{\text{OUT}}{v_x} \left( \rho U + \frac{1}{2} \rho v^2 \right) \Big|_{x+\Delta x} \right\} \\ & + \overset{(y \text{ FACE})}{\Delta x \Delta z} \left\{ v_y \left( \rho U + \frac{1}{2} \rho v^2 \right) \Big|_y - v_y \left( \rho U + \frac{1}{2} \rho v^2 \right) \Big|_{y+\Delta y} \right\} \\ & + \Delta x \Delta y \left\{ v_z \left( \rho U + \frac{1}{2} \rho v^2 \right) \Big|_z - v_z \left( \rho U + \frac{1}{2} \rho v^2 \right) \Big|_{z+\Delta z} \right\} \end{aligned}$$

COND.

NET RATE OF ENERGY IN BY COND.

$$\Delta y \Delta z \left\{ q_x|_x - q_x|_{x+\Delta x} \right\} + \Delta x \Delta z \left\{ q_y|_y - q_y|_{y+\Delta y} \right\} \\ + \Delta x \Delta y \left\{ q_z|_z - q_z|_{z+\Delta z} \right\}$$

WORK DONE 

AGAINST VOLUMETRIC FORCE (BODY)  
(GRAVITY) ✓

AGAINST SURFACE FORCES  
↓  
(PRESSURE, VISCOS) ✓

RATE OF WORK DONE = FORCE X VEL. IN THE  
DIRECTN OF FORCE

RATE <sup>OF</sup> WORK DONE  
AGAINST GRAVITATIONAL FORCE

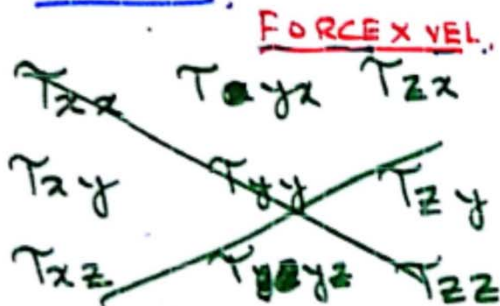
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$$- \rho(\Delta x \Delta y \Delta z)(v_x g_x + v_y g_y + v_z g_z)$$

RATE OF WORK DONE  
AGAINST STATIC PR.

$$\Delta y \Delta z \left\{ \rho v_x \Big|_{x+\Delta x} - \rho v_x \Big|_x \right\} + \Delta x \Delta z \left\{ \rho v_y \Big|_{y+\Delta y} - \rho v_y \Big|_y \right\} + \Delta x \Delta y \left\{ \rho v_z \Big|_{z+\Delta z} - \rho v_z \Big|_z \right\}$$

AGAINST VISCOUS  
FORCES



$\Delta x \Delta z$

$$\begin{Bmatrix} \tau_{xx} v_x \\ \tau_{xy} v_y \\ \tau_{xz} v_z \end{Bmatrix} \quad \begin{Bmatrix} \tau_{yx} v_x \\ \tau_{yy} v_y \\ \tau_{yz} v_z \end{Bmatrix} \quad \begin{Bmatrix} \tau_{zx} v_x \\ \tau_{zy} v_y \\ \tau_{zz} v_z \end{Bmatrix}$$

$\Delta x \Delta y$

NET  
RATE

$$\Delta y \Delta z \left\{ (\tau_{xx} v_x + \tau_{xy} v_y + \tau_{xz} v_z) \Big|_{x+\Delta x} - (\tau_{xx} v_x + \dots) \Big|_x \right. \\ \left. + 2 \tau'_{xy} + 2 \tau'_{yz} \right\}$$

$$\boxed{\rho c \frac{DT}{Dt}} = k \nabla^2 T - T \left( \frac{\partial P}{\partial T} \right) \nabla \cdot \mathbf{v} + \mu \phi_v$$

$\uparrow$  HEAT COND.       $\uparrow$  EXPANSION EFFECTS       $\uparrow$  VISCOUS DISSIPATION.

$\rho c \left[ \frac{\partial T}{\partial t} + v_x \frac{\partial T}{\partial x} + v_y \frac{\partial T}{\partial y} \dots \right]$

$k \left( \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \dots \right)$

SIMPLIFICATIONS

$\rho \neq f(T)$  , NO  $\phi$

$$\Rightarrow \rho C_p \frac{DT}{Dt} = k \nabla^2 T + \dot{q}$$

BIRD  
STEWART  
LIGHTFOOT

SOLID  $\rho$  CONST

$$\rho C_p \frac{DT}{Dt} = k \nabla^2 T + \dot{q}$$

$$\rho C_p \frac{\partial T}{\partial t} = k \nabla^2 T + \dot{q}$$

COND.  
EQN.

## Heat Transfer Fundamentals

Fourier's Law  $q_x = -k A \frac{dT}{dx}$     Newton's Law of Cooling  $q = h A (T_s - T_\infty)$

Equation of Energy in Cartesian and Cylindrical Coordinate Systems

$$\rho \hat{C}_p \left( \frac{\partial T}{\partial t} + \underbrace{v_x \frac{\partial T}{\partial x} + v_y \frac{\partial T}{\partial y} + v_z \frac{\partial T}{\partial z}}_{\text{Convection}} \right) = k \left[ \underbrace{\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}}_{\text{Conduction}} \right] + \underbrace{\mu \phi_v}_{\text{Viscous Dissipation}} + \underbrace{\dot{Q}}_{\text{Generation}}$$

Transient,    Convection,    Conduction,    Viscous Dissipation,    Generation

$$\rho \hat{C}_p \left( \frac{\partial T}{\partial t} + v_r \frac{\partial T}{\partial r} + \frac{v_\theta}{r} \frac{\partial T}{\partial \theta} + v_z \frac{\partial T}{\partial z} \right) = k \left[ \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 T}{\partial \theta^2} + \frac{\partial^2 T}{\partial z^2} \right] + \mu \phi_v + \dot{Q}$$



## Equations of Energy

*Cartesian coordinates (x, y, z)*

$$\rho \hat{C}_p \left( \frac{\partial T}{\partial t} + v_x \frac{\partial T}{\partial x} + v_y \frac{\partial T}{\partial y} + v_z \frac{\partial T}{\partial z} \right) = k \left[ \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right] + \mu \phi_v + \dot{Q}$$

*Cylindrical coordinates (r,  $\theta$ , z)*

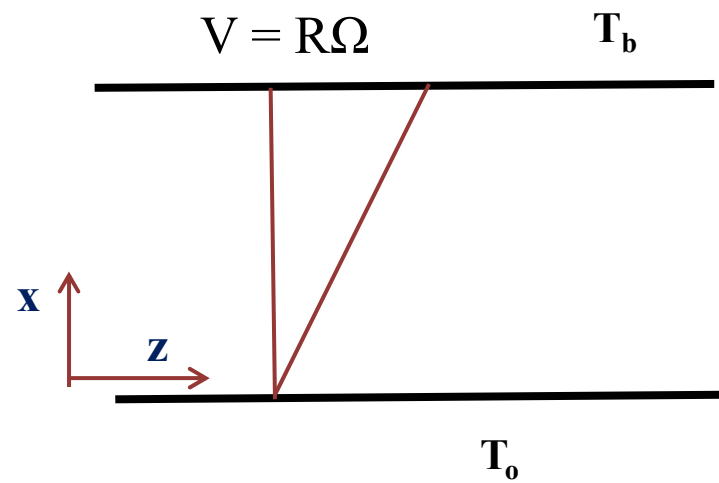
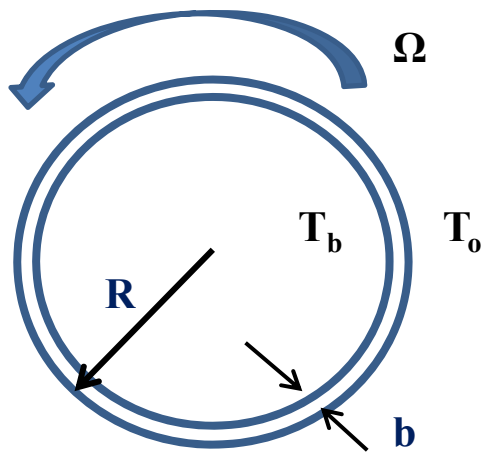
$$\rho \hat{C}_p \left( \frac{\partial T}{\partial t} + v_r \frac{\partial T}{\partial r} + \frac{v_\theta}{r} \frac{\partial T}{\partial \theta} + v_z \frac{\partial T}{\partial z} \right) = k \left[ \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 T}{\partial \theta^2} + \frac{\partial^2 T}{\partial z^2} \right] + \mu \phi_v + \dot{Q}$$

*Spherical coordinates (r,  $\theta$ ,  $\phi$ )*

$$\rho \hat{C}_p \left( \frac{\partial T}{\partial t} + v_r \frac{\partial T}{\partial r} + \frac{v_\theta}{r} \frac{\partial T}{\partial \theta} + \frac{v_\phi}{r \sin \theta} \frac{\partial T}{\partial \phi} \right) = k \left[ \frac{1}{r^2} \frac{\partial}{\partial r} \left( r^2 \frac{\partial T}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial T}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 T}{\partial \phi^2} \right] + \mu \phi_v + \dot{Q}$$

## Viscous Heat Generation

Heat Conduction with Viscous Heat Source



## The Dissipation Function

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*Cartesian coordinates (x, y, z):*

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$$\Phi_v = 2 \left[ \left( \frac{\partial v_x}{\partial x} \right)^2 + \left( \frac{\partial v_y}{\partial y} \right)^2 + \left( \frac{\partial v_z}{\partial z} \right)^2 \right] + \left[ \frac{\partial v_y}{\partial x} + \frac{\partial v_x}{\partial y} \right]^2 + \left[ \frac{\partial v_z}{\partial y} + \frac{\partial v_y}{\partial z} \right]^2 + \left[ \frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x} \right]^2 - \frac{2}{3} \left[ \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right]^2$$

---

*Cylindrical coordinates (r,  $\theta$ , z):*

---

$$\Phi_v = 2 \left[ \left( \frac{\partial v_r}{\partial r} \right)^2 + \left( \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} \right)^2 + \left( \frac{\partial v_z}{\partial z} \right)^2 \right] + \left[ r \frac{\partial}{\partial r} \left( \frac{v_\theta}{r} \right) + \frac{1}{r} \frac{\partial v_r}{\partial \theta} \right]^2 + \left[ \frac{1}{r} \frac{\partial v_z}{\partial \theta} + \frac{\partial v_\theta}{\partial z} \right]^2 + \left[ \frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} \right]^2 - \frac{2}{3} \left[ \frac{1}{r} \frac{\partial}{\partial r} (rv_r) + \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{\partial v_z}{\partial z} \right]^2$$

$$b \frac{d^2 T}{dx^2} = -\mu \left( \frac{dv_z}{dx} \right)^2 \quad \left| \quad v_z = \left( \frac{x}{b} \right) v \right.$$

$$\begin{aligned} x=0 & \quad T=T_0 \\ x=b & \quad T=T_b \end{aligned}$$

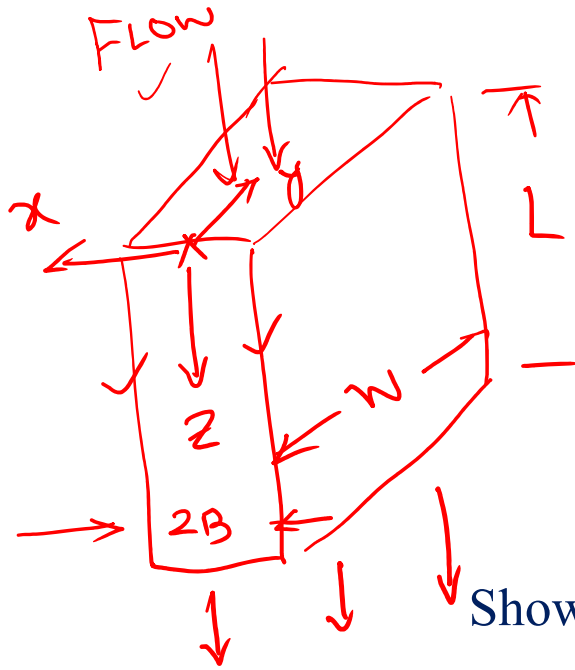
$$\frac{T-T_0}{T_b-T_0} = \left( \frac{x}{b} \right) + \frac{1}{2} Br \left( \frac{x}{b} \right) \left[ 1 - \frac{x}{b} \right]$$

$$\boxed{Br > 2}$$

THERE IS A MAX T AT  
AN INTERMEDIATE POSITION

BRINKMAN No.

$$Br = \frac{\mu v^2 / L}{b(T_b - T_0) / L}$$



LAMINAR FLOW IN A SLIT

$$W \gg 2B \quad v_z = f(x) \text{ ONLY} \\ \neq f(z, y)$$

BOTH PLATES AT  $T_0$

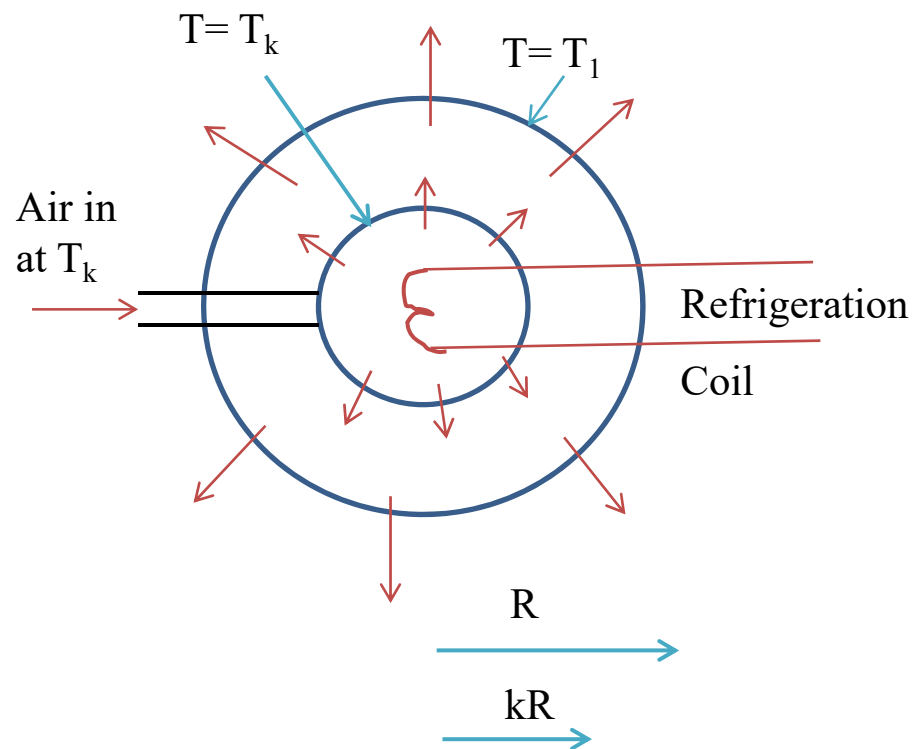
$$T = f(x), \quad v_{\max}$$

Show that the temperature profile will be of the following form

$$\checkmark \quad T - T_0 = \frac{1}{3} \frac{\mu (v_{\max})^2}{k} \left[ 1 - \left( \frac{x}{B} \right)^4 \right]$$



## Transpiration Cooling



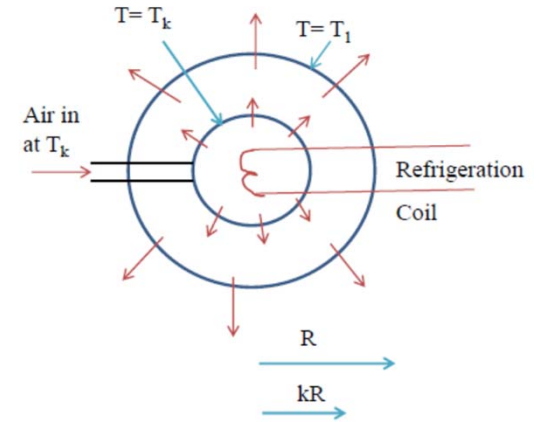
Two concentric porous spherical shells.  $T_k$  is to be kept lower than  $T_1$  by passing dry air at  $T_k$  in a radially outward direction from the inner shell and through the outer one to the outside.

Find expression for the rate of heat removal from the inner sphere as a function of the mass flow rate of the air at SS condition.

Here,  $v_\theta = 0$ ,  $v_\phi = 0$  and the equation of continuity becomes -

$$\frac{1}{r^2} \frac{d}{dr} (r^2 \rho v_r) = 0 \Rightarrow r^2 \rho v_r = C = \frac{W_r}{4\pi}$$

$$v_r = \frac{W_r}{4\pi r^2 \rho}$$



$W_r$  is the radial mass flow rate of the gas

Equation of energy

Spherical coordinates  $(r, \theta, \phi)$

$$\rho \hat{C}_p \left( \frac{\partial T}{\partial t} + v_r \frac{\partial T}{\partial r} + \frac{v_\theta}{r} \frac{\partial T}{\partial \theta} + \frac{v_\phi}{r \sin \theta} \frac{\partial T}{\partial \phi} \right) = k \left[ \frac{1}{r^2} \frac{\partial}{\partial r} \left( r^2 \frac{\partial T}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial T}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 T}{\partial \phi^2} \right] + \mu \phi_v + \dot{Q}$$



Governing Equation

$$\rho \hat{C}_p \left( v_r \frac{dT}{dr} \right) = k \left[ \frac{1}{r^2} \frac{d}{dr} \left( r^2 \frac{dT}{dr} \right) \right]$$

Using Equation of Continuity

$$\frac{dT}{dr} = \frac{4 \pi k}{W_r \hat{C}_p} \frac{d}{dr} \left( r^2 \frac{dT}{dr} \right)$$

Integrating

$$\frac{T - T_1}{T_k - T_1} = \frac{e^{-R_o/r} - e^{-R_o/R}}{e^{-R_o/kR} - e^{-R_o/R}} \quad ; \quad R_o = \frac{W_r \hat{C}_p}{4 \pi k}$$

For small  $R_o$

$$\frac{T - T_1}{T_k - T_1} = \frac{1/r - 1/R}{1/kr - 1/R}$$

If  $R$  is infinite and  $kR$  is finite

$$\frac{T - T_\infty}{T_k - T_\infty} = \frac{kR}{r}$$

The heat flow to the inner sphere in the  $-r$  direction

$$Q = -4\pi (\kappa R)^2 q_r \Big|_{r=kR} = 4\pi k (\kappa R)^2 \frac{dT}{dr} \Big|_{r=kR}$$

$$Q = 4\pi k R \frac{(T_1 - T_\kappa)}{\left[ e^{\left(\frac{R_o}{\kappa R}\right)^{(1-\kappa)}} - 1 \right]}$$

When  $R_o = 0$ , the rate of heat removal at zero gas rate

$$Q_o = 4\pi kR \frac{(T_1 - T_\kappa)}{(1 - \kappa)}$$

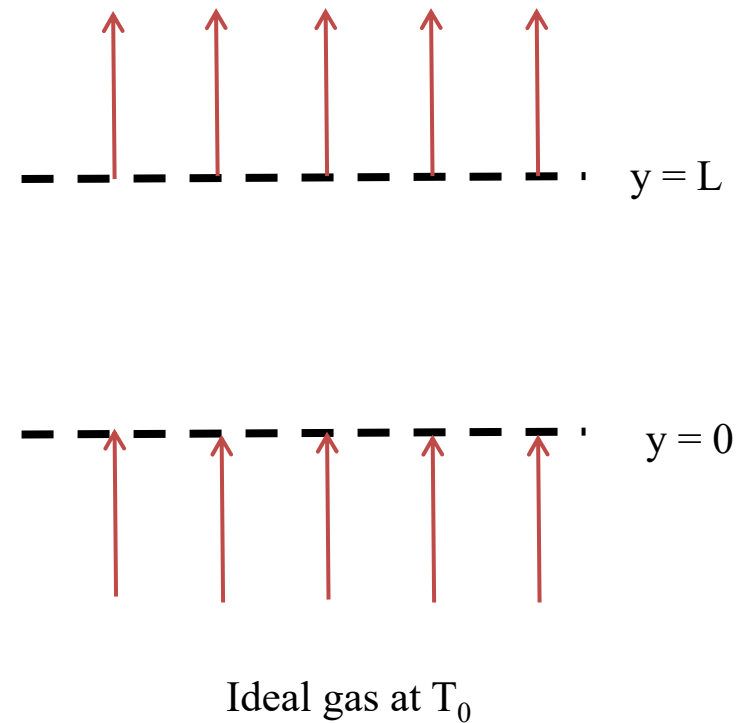
The effectiveness  $\epsilon$  of the transpiration is given by  $\epsilon = \frac{Q_o - Q}{Q_o} = 1 - \frac{\phi}{e^\phi - 1}$  ;  $\phi = \frac{R_o(1 - \kappa)}{\kappa R}$

For low values of  $\phi$ ,  $\epsilon$  tends to  $\phi/2$

**Transpiration can be a very effective method for reducing the heat transfer rates.  
Especially important where heat and mass transfer occur simultaneously.**

## Transpiration Cooling

Two large porous horizontal plates are separated by a small distance  $L$ . The upper plate is at a temperature of  $T = T_L$  while the lower plate is to be maintained at a lower temperature  $T_0$ . To reduce the amount of heat that must be removed from the lower plate, an ideal gas at  $T_0$  is blown upward through both plates at a steady rate. Develop an expression of temperature distribution and the amount of heat  $q_0$  that must be removed from the cold plate per unit area as a function of the fluid properties and the flow rate of the ideal gas.



$$\rho \hat{C}_p \left( \frac{\partial T}{\partial t} + v_x \frac{\partial T}{\partial x} + v_y \frac{\partial T}{\partial y} + v_z \frac{\partial T}{\partial z} \right) = k \left[ \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right] + \mu \phi_v + \dot{Q}$$



SS, 1D conduction, convection, no  
viscous dissipation, no heat generation

$$\rho \hat{C}_p \left( v_y \frac{dT}{dy} \right) = k \left[ \frac{d^2 T}{dy^2} \right]$$

$$\frac{d^2 T}{dy^2} - \frac{1}{Y_o} \frac{dT}{dY} = 0; \quad Y_o = \frac{k}{\rho \hat{C}_p v_y}$$

To evaluate  $v_y$  equation of continuity may be used.

$$\frac{\partial \rho v_x}{\partial x} + \frac{\partial \rho v_y}{\partial y} + \frac{\partial \rho v_z}{\partial z} = 0$$

$$\rho v_y = \text{Const} \quad \dot{m} = \text{mass flow rate, } A = \text{area}$$

The governing equation can then be solved.

$$\frac{d^2 T}{dy^2} - \frac{1}{Y_o} \frac{dT}{dY} = 0; \quad Y_o = \frac{k}{\rho \widehat{C_p} v_y}$$

With the boundary conditions

$$T = T_o \quad \text{at} \quad y = 0$$

$$T = T_L \quad \text{at} \quad y = L$$

To obtain

$$\frac{T - T_L}{T_o - T_L} = \frac{e^{\left(\frac{y\phi}{L}\right)} - e^\phi}{1 - e^\phi}$$

Rate of heat removal by Fourier's law

$$q_o = k \left. \frac{dT}{dy} \right|_{y=0} = \frac{k(T_L - T_o)}{L} \frac{\phi}{e^\phi - 1}$$

## The Dissipation Function

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*Cartesian coordinates (x, y, z):*

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$$\Phi_v = 2 \left[ \left( \frac{\partial v_x}{\partial x} \right)^2 + \left( \frac{\partial v_y}{\partial y} \right)^2 + \left( \frac{\partial v_z}{\partial z} \right)^2 \right] + \left[ \frac{\partial v_y}{\partial x} + \frac{\partial v_x}{\partial y} \right]^2 + \left[ \frac{\partial v_z}{\partial y} + \frac{\partial v_y}{\partial z} \right]^2 + \left[ \frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x} \right]^2 - \frac{2}{3} \left[ \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right]^2$$

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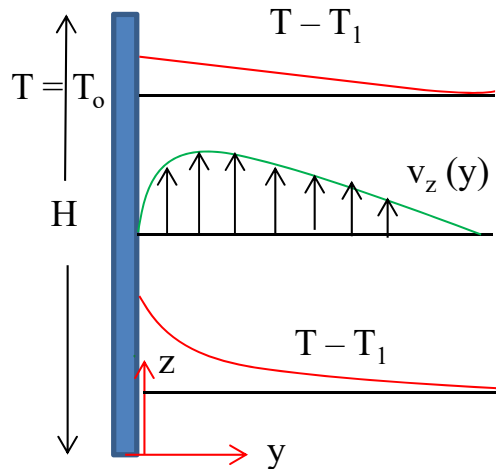
*Cylindrical coordinates (r,  $\theta$ , z):*

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$$\Phi_v = 2 \left[ \left( \frac{\partial v_r}{\partial r} \right)^2 + \left( \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} \right)^2 + \left( \frac{\partial v_z}{\partial z} \right)^2 \right] + \left[ r \frac{\partial}{\partial r} \left( \frac{v_\theta}{r} \right) + \frac{1}{r} \frac{\partial v_r}{\partial \theta} \right]^2 + \left[ \frac{1}{r} \frac{\partial v_z}{\partial \theta} + \frac{\partial v_\theta}{\partial z} \right]^2 + \left[ \frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} \right]^2 - \frac{2}{3} \left[ \frac{1}{r} \frac{\partial}{\partial r} (rv_r) + \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{\partial v_z}{\partial z} \right]^2$$



## Free Convection from a Vertical Plate



Flat plate heated to a temperature of  $T_0$  and is suspended in a large body of fluid. Which is at temperature  $T_1$ . Fluid near the plate rises due to buoyancy.

Deduce the dependence of the heat loss on system variables.  
Free convection form of the equation of motion is to be used.

Assumptions:

1. Dimension in the x direction is very large; thus  $v_y$  and  $v_z$  are functions of y and z only
2.  $v_y$  is small; fluid almost moves directly upward and therefore, y component of the equation of motion can be neglected.

$$\beta = \frac{1}{V} \frac{\Delta V}{\Delta T} \quad \text{Buoyant Force} = \rho g \Delta V = \rho g \beta V \Delta T \quad ; \quad \text{Buoyant Force} / \text{Volume} = \rho g \beta \Delta T$$

## The Boussinesq Equation of Motion for Forced and Free Convection

In non-isothermal flow, the fluid  $\rho$  and  $\mu$  depend on  $T$  as well as  $P$ . The variation in  $\rho$  is particularly important as it gives rise to buoyant forces, and thus to free convection.

The buoyant force appears automatically when an equation of state is inserted into the equation of motion. For example, one can use the simplified equation of state into the equation of motion (known as the Boussinesq approximation).

$$\rho(T) = \bar{\rho} - \bar{\rho} \bar{\beta} (T - \bar{T}); \quad \text{Where} \quad \beta = - \left( \frac{1}{\rho} \right) (\partial \rho / \partial T)_p, \text{ evaluated at } \bar{T}$$

When the above equation is substituted into the  $\rho g$  term of the Eq. of motion, but not into the  $\rho D\mathbf{v}/Dt$  term), one would get the **Boussinesq equation**

$$\rho \frac{D\mathbf{v}}{Dt} = \left( -\cancel{\nabla P} + \bar{\rho} g \right) - [\nabla \cdot \boldsymbol{\tau}] - \cancel{\bar{\rho} g \bar{\beta} (T - \bar{T})}$$

Neglected in Free Convection

Neglected in Forced Convection

Continuity Equation

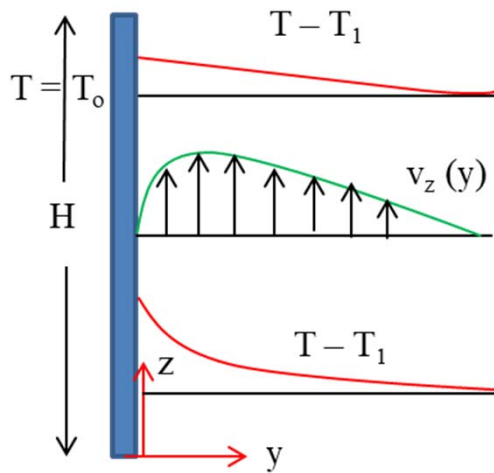
$$\frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 0$$

Equation of  
Motion

$$\rho \left( v_y \frac{\partial v_z}{\partial y} + v_z \frac{\partial v_z}{\partial z} \right) = \mu \left[ \frac{\partial^2 v_z}{\partial y^2} + \frac{\partial^2 v_z}{\partial z^2} \right] + \rho g \beta (T - T_1)$$

Equation of  
Energy

$$\rho \hat{C}_p \left( v_y \frac{\partial}{\partial y} + v_z \frac{\partial}{\partial z} \right) (T - T_1) = k \left[ \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right] (T - T_1)$$



## Boundary Conditions

$$y = 0, \quad v_y = v_z = 0, \quad T = T_o$$

$$y \rightarrow \infty, \quad v_z = 0, \quad T = T_1$$

$$z \rightarrow -\infty, \quad v_y = v_z = 0, \quad T = T_1$$

$$\rho \left( v_y \frac{\partial v_z}{\partial y} + v_z \frac{\partial v_z}{\partial z} \right) = \mu \left[ \frac{\partial^2 v_z}{\partial y^2} + \frac{\partial^2 v_z}{\partial z^2} \right] + \rho g \beta (T - T_1)$$

$$\rho \hat{C}_p \left( v_y \frac{\partial}{\partial y} + v_z \frac{\partial}{\partial z} \right) (T - T_1) = k \left[ \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right] (T - T_1)$$

## Dimensionless variables

$$\theta = \frac{T - T_1}{T_o - T_1} \quad \zeta = z/H$$

$$\eta = \left( \frac{B}{\mu \alpha H} \right)^{1/4} y$$

$$\phi_z = \left( \frac{\mu}{B \alpha H} \right)^{1/4} v_z$$

$$\phi_y = \left( \frac{\mu H}{\alpha^3 B} \right)^{1/4} v_y$$

$$\phi_y \text{ \& } \phi_z = f(\eta, \zeta, \text{Pr})$$

$$\theta = f(\eta, \zeta, \text{Pr})$$

## Governing Equations

$$\frac{\partial \phi_y}{\partial \eta} + \frac{\partial \phi_z}{\partial \zeta} = 0$$

$$\frac{1}{\text{Pr}} \left( \phi_y \frac{\partial \phi_z}{\partial \eta} + \phi_z \frac{\partial \phi_y}{\partial \zeta} \right) = \frac{\partial^2 \phi_z}{\partial \eta^2} + \phi$$

$$\left( \phi_y \frac{\partial \theta}{\partial \eta} + \phi_z \frac{\partial \theta}{\partial \zeta} \right) = \frac{\partial^2 \theta}{\partial \eta^2}$$

$$q_{avg} = + \frac{k}{H} \int_0^H - \frac{\partial T}{\partial y} \Big|_{y=0} dz$$

$$q_{avg} = k(T_o - T_1) \left( \frac{B}{\mu \alpha H} \right) \int_0^1 - \frac{\partial \theta}{\partial \eta} \Big|_{\eta=0} d\zeta$$

$$\theta = f(\eta, \zeta, \text{Pr})$$

$$q_{avg} = C \frac{k}{H} (T_o - T_1) (Gr \cdot \text{Pr})^{1/4}$$

$$C = f(\text{Pr}) \rightarrow \text{Strong / Weak Function}$$

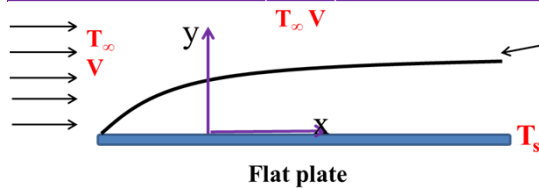
Experimental measurements provide, in laminar flow range ( $GrPr < 10^9$ ), variation of  $C$  with  $Pr$  is indeed quite small.

| Pr | 0.73  | 10    | $10^2$ | $10^3$ |
|----|-------|-------|--------|--------|
| C  | 0.517 | 0.612 | 0.652  | 0.653  |

## Convection Boundary Layers

Velocity boundary layer - Fundamental importance to convection transport problems - Concept of  $C_f$

Thermal boundary layer - BL thickness is the value for which the ratio



$$\frac{T_s - T}{T_s - T_\infty} \approx 0.99$$

Concept of convection coefficient

At any distance  $x$  from the leading edge, the local heat flux may be obtained by applying Fourier's law to the fluid at  $y = 0$



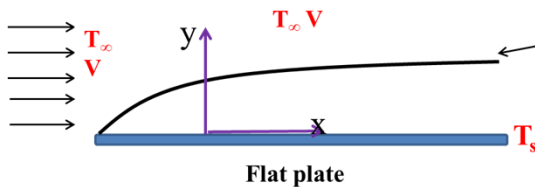
$$q_s = -k \left. \frac{\partial T}{\partial y} \right|_{y=0} = h(T_s - T_\infty)$$

$$h = \frac{-k_f \left. \frac{\partial T}{\partial y} \right|_{y=0}}{(T_s - T_\infty)}$$

If  $T^* \equiv \frac{T - T_s}{T_\infty - T_s}$  and  $x^* \equiv x / L$

$$\frac{hL}{k_f} = \left. \frac{\partial T^*}{\partial y^*} \right|_{y=0}$$

- Nusselt number is the dimensionless temp gradient at the interface
- Conditions in the thermal BL strongly influences and determine the rate of heat transfer across the BL
- $(T_s - T_\infty)$  is a constant but  $\delta_t$  increases with  $x$ , therefore, the gradient decreases with increasing  $x$  and  $q_s$  and  $h$  decrease with  $x$



|                          | <b>Velocity B.L.</b>              | <b>Thermal B.L.</b>                       |
|--------------------------|-----------------------------------|-------------------------------------------|
| Characterized by         | Velocity Gradient<br>Shear Stress | Temperature gradient<br>Heat transfer     |
| Principal manifestations | Surface friction                  | Convection heat transfer                  |
| Key parameters           | Friction coefficient, $C_f$       | Convective heat transfer coefficient, $h$ |

## Transfer Equations

**Momentum Eqn**

$$v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} = -\frac{1}{\rho} \frac{\partial P}{\partial x} + \gamma \frac{\partial^2 v_x}{\partial y^2}$$

**Thermal Energy Eqn**

$$\rho \hat{C}_p \left( v_x \frac{\partial T}{\partial x} + v_y \frac{\partial T}{\partial y} \right) = k \left[ \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right] + \mu \phi_v + \dot{Q}$$

**Numerical solution of M eqn.**  $\delta = \frac{5.0 x}{\sqrt{\text{Re}_x}} \quad \tau_s = \frac{0.332 \rho U^2}{\sqrt{\text{Re}_x}} \quad C_{f,x} = 0.664 \text{Re}_x^{-1/2}$

**The energy equation can now be solved**

## Thermal Energy Eqn

$$\rho \hat{C}_p \left( v_x \frac{\partial T}{\partial x} + v_y \frac{\partial T}{\partial y} \right) = k \left[ \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right] + \mu \phi_v + \dot{Q}$$

$$f(\eta) \equiv \frac{\psi}{\sqrt{\nu x U_\infty}} \quad \eta \equiv y \sqrt{\frac{U_\infty}{\nu x}} \quad T^* \equiv \frac{T - T_s}{T_\infty - T_s} \quad 2 \frac{d^3 f}{d\eta^3} + f \frac{d^2 f}{d\eta^2} = 0$$

Dimensionless form of  
the resulting ODE

$$\frac{d^2 T^*}{d\eta^2} + \frac{\text{Pr}}{2} \underbrace{f}_{\text{Dependence of the thermal solution on the hydrodynamic conditions}} \frac{dT^*}{d\eta} = 0$$

Dependence of  
the thermal  
solution on the  
hydrodynamic  
conditions

Boundary conditions

$$\begin{aligned} T^* &= 0 \quad \text{at} \quad \eta = 0 & T^* &= \frac{T - T_s}{T_\infty - T_s} \\ T^* &= 1 \quad \text{at} \quad \eta = \infty \end{aligned}$$

The energy equation was solved numerically for different values of Pr.

For  $0.6 \leq \text{Pr} \leq 50$ , results for the surface temperature gradient  $\left. \frac{\partial T}{\partial y} \right|_{y=0}$  can be expressed as

$$\left. \frac{dT^*}{d\eta} \right|_{\eta=0} = 0.332 \text{Pr}^{1/3}$$

Local convection  
coefficient

$$h_x = \frac{q_s''}{T_s - T_\infty} = -\frac{1}{T_s - T_\infty} \frac{\partial T}{\partial y} \bigg|_{y=0} = \frac{k}{L} \frac{\partial T^*}{\partial y^*} \bigg|_{y=0} = k \left( \frac{U_\infty}{\nu x} \right)^{1/2} \frac{dT^*}{d\eta^*} \bigg|_{\eta=0}$$

$$Nu_x = \frac{h_x x}{k} = 0.332 \text{Re}_x^{1/2} \text{Pr}^{1/3} \quad \begin{array}{l} 0.6 \leq \text{Pr} \leq 50 \\ \text{Re} < 5 \times 10^5 \end{array}$$

From the solution of the governing equations

$$\frac{\delta}{\delta_t} \approx \text{Pr}^{1/3} \quad \text{Re}_{x,c} = 5 \times 10^5$$

### Average Coefficients

$$\overline{C}_{f,x} = \frac{\overline{\tau_{s,x}}}{\rho U_\infty^2 / 2}; \quad \overline{\tau_{sx}} = \frac{1}{x} \int_0^x \overline{\tau_{s,x}} dx = \frac{1}{x} \int_0^x 0.332 \frac{\rho U_\infty^2}{\sqrt{\text{Re}_x}} dx$$

$$\overline{C}_{f,x} = 1.328 \text{Re}_x^{-1/2} = 2 C_{f,x}$$

$$\bar{h}_x = \frac{1}{x} \int_0^x h_x dx = 0.332 \left( \frac{k}{x} \right) \text{Pr}^{1/3} \left( \frac{U_\infty}{\nu} \right)^{1/2} \int_0^x \frac{dx}{x^{1/2}} = 2 h_x$$

$$\overline{Nu}_x = \frac{\bar{h}_x x}{k} = 0.664 \text{Re}_x^{1/2} \text{Pr}^{1/3} \quad \begin{array}{l} \text{Re}_{x,c} = 5 \times 10^5 \\ 0.6 \leq \text{Pr} \leq 50 \end{array}$$

## Liquid Metals in Boundary Layer Flow & Heat Transfer

Consider a liquid metal (  $Pr \ll 1$  ), with the freestream conditions  $U_\infty$  and  $T_\infty$  , in parallel flow over an isothermal flat plate at  $T_s$  . Simplify the boundary layer energy equation (given below) for this special case and apply appropriate initial and boundary conditions to solve for the boundary layer temperature field  $T(x, y)$ . Use the result to obtain an expression for the local Nusselt number as a function of Peclet number.

$$v_x \frac{\partial T}{\partial x} + v_y \frac{\partial T}{\partial y} = \alpha \left( \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right)$$



$$v_x \frac{\partial T}{\partial x} + v_y \frac{\partial T}{\partial y} = \alpha \left( \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right)$$

$$U \frac{\partial T}{\partial x} = \alpha \frac{\partial^2 T}{\partial y^2}$$

i)  $V_x = U$  everywhere as  $\delta_L \gg \delta$

ii)  $\frac{\partial T}{\partial y}$  is small

iii) Conductive transport is in the y-direction

Growth of BL for Liq.  
Metals

$$U \frac{\partial T}{\partial x} = \alpha \frac{\partial^2 T}{\partial y^2}$$

$$T^* = \frac{T - T_\infty}{T_s - T_\infty}$$

IC:  $x = 0$ ;  $T^* = 0$  for all  $y$

BC:  $y = 0$ ;  $T^* = 1$  for all  $x$

$y = \infty$ ;  $T^* = 0$  for all  $x$

$$\frac{\partial T^*}{\partial x} = \frac{\alpha}{U} \frac{\partial^2 T^*}{\partial y^2}$$

Standard error function  
solution

$$\frac{T_s - T}{T_s - T_\infty} = T^* = 1 - \operatorname{erf} \left( \frac{y}{\sqrt{4 \frac{\alpha}{U} x}} \right)$$

At the L-S interface

$$-k \left. \frac{\partial T}{\partial y} \right|_{y=0} = h (T_s - T_\infty)$$

For error function

$$\left. \frac{\partial}{\partial y} \left( \operatorname{erf} \frac{y}{\sqrt{4a x}} \right) \right|_{y=0} = \frac{1}{\sqrt{\pi a x}}$$

$$Nu_x = 0.565 (\text{Re}_x \text{Pr})^{1/2} = 0.565 (Pe)^{1/2}$$

Useful relation for liquid metals

